SPECIAL DIRECT SYSTEMS BUILT FROM LEFT RESOLUTIONS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Left resolution of complexes. lemma-subcategory-left-resolution (same chapter, carded in the old generation). Supplies the left decomposition engine applied twice in the inductive construction.
- Improving triangle morphisms up to homotopy. lemma-improve-distinguished-triangle-homotopy (same chapter, carded in the old generation). Supplies the homotopy-improvement engine used to glue the triangle at each stage.
- The homotopy category is triangulated. proposition-homotopy-category-triangulated (same chapter, carded in the old generation). Supplies the triangulated-category bookkeeping used to assemble the special direct system.

Source reference. Original lemma 0G6C: sources/stacks-derived/pinned-original-source.tex, source lines 3676–3676. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05T7: sources/stacks-derived/pinned-original-source.tex, source lines 5519–5519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 014S: sources/stacks-derived/pinned-original-source.tex, source lines 3537–3537. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{P} \subset \mathrm{Ob}(\mathcal{A})$ be a subset. Assume $\mathcal{P}$ contains 0, is closed under (finite) direct sums, and every object of $\mathcal{A}$ is a quotient of an element of $\mathcal{P}$. Let $K^\bullet$ be a complex. There exists a commutative diagram*

$$\begin{array}{ccccc} P_1^\bullet & \longrightarrow & P_2^\bullet & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \\ \tau_{\leq 1}K^\bullet & \longrightarrow & \tau_{\leq 2}K^\bullet & \longrightarrow & \ldots \end{array}$$

in the category of complexes such that

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *the vertical arrows are quasi-isomorphisms and termwise surjective,*
(2) $P_n^\bullet$ *is a bounded above complex with terms in* $\mathcal{P}$*,*
(3) *the arrows* $P_n^\bullet \to P_{n+1}^\bullet$ *are termwise split injections and each cokernel* P_{n+1}^i/P_n^i *is an element of* $\mathcal{P}$*.*

Proof. We are going to use that the homotopy category $K(\mathcal{A})$ is a triangulated category, see Proposition 014S. By Lemma 05T7 we can find a termwise surjective map of complexes $P_1^\bullet \to \tau_{\leq 1}K^\bullet$ which is a quasi-isomorphism such that the terms of $P_1^\bullet$ are in $\mathcal{P}$. By induction it suffices, given $P_1^\bullet, \ldots, P_n^\bullet$ to construct $P_{n+1}^\bullet$ and the maps $P_n^\bullet \to P_{n+1}^\bullet$ and $P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$.

Choose a distinguished triangle $P_n^\bullet \to \tau_{\leq n+1}K^\bullet \to C^\bullet \to P_n^\bullet[1]$ in $K(\mathcal{A})$. Applying Lemma 05T7 we choose a map of complexes $Q^\bullet \to C^\bullet$ which is a quasi-isomorphism such that the terms of $Q^\bullet$ are in $\mathcal{P}$. By the axioms of triangulated categories we may fit the composition $Q^\bullet \to C^\bullet \to P_n^\bullet[1]$ into a distinguished triangle $P_n^\bullet \to P_{n+1}^\bullet \to Q^\bullet \to P_n^\bullet[1]$ in $K(\mathcal{A})$. By Lemma 0G6C we may and do assume $0 \to P_n^\bullet \to P_{n+1}^\bullet \to Q^\bullet \to 0$ is a termwise split short exact sequence. This implies that the terms of $P_{n+1}^\bullet$ are in $\mathcal{P}$ and that $P_n^\bullet \to P_{n+1}^\bullet$ is a termwise split injection whose cokernels are in $\mathcal{P}$. By the axioms of triangulated categories we obtain a map of distinguished triangles

$$\begin{array}{ccccccc} P_n^\bullet & \longrightarrow & P_{n+1}^\bullet & \longrightarrow & Q^\bullet & \longrightarrow & P_n^\bullet[1] \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ P_n^\bullet & \longrightarrow & \tau_{\leq n+1}K^\bullet & \longrightarrow & C^\bullet & \longrightarrow & P_n^\bullet[1] \end{array}$$

in the triangulated category $K(\mathcal{A})$. Choose an actual morphism of complexes $f : P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$. The left square of the diagram above commutes up to homotopy, but as $P_n^\bullet \to P_{n+1}^\bullet$ is a termwise split injection we can lift the homotopy and modify our choice of f to make it commute. Finally, f is a quasi-isomorphism, because both $P_n^\bullet \to P_n^\bullet$ and $Q^\bullet \to C^\bullet$ are.

At this point we have all the properties we want, except we don't know that the map $f : P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$ is termwise surjective. Since we have the commutative diagram

$$\begin{array}{ccc} P_n^\bullet & \longrightarrow & P_{n+1}^\bullet \\ \downarrow & & \downarrow \\ \tau_{\leq n}K^\bullet & \longrightarrow & \tau_{\leq n+1}K^\bullet \end{array}$$

of complexes, by induction hypothesis we see that f is surjective on terms in all degrees except possibly n and $n+1$. Choose an object $P \in \mathcal{P}$ and a surjection $q : P \to K^n$. Consider the map

$$g : P^\bullet = (\ldots \to 0 \to P \xrightarrow{1} P \to 0 \to \ldots) \longrightarrow \tau_{\leq n+1}K^\bullet$$

with first copy of P in degree n and maps given by q in degree n and $d_K \circ q$ in degree $n+1$. This is a surjection in degree n and the cokernel in degree $n+1$ is $H^{n+1}(\tau_{\leq n+1}K^\bullet)$; to see this recall that $\tau_{\leq n+1}K^\bullet$ has $\mathrm{Ker}(d_K^{n+1})$ in degree $n+1$. However, since f is a quasi-isomorphism we know that $H^{n+1}(f)$ is surjective. Hence after replacing $f : P_{n+1}^\bullet \to \tau_{\leq n+1}K^\bullet$ by $f \oplus g : P_{n+1}^\bullet \oplus P^\bullet \to \tau_{\leq n+1}K^\bullet$ we win. $\square$